



Applied Mathematics and Informatics Program

Earthquake Magnitude Estimation from Early P-Wave Observations Using a Machine Learning Approach

Papakov Aidarbek

A Thesis Submitted to the Applied Mathematics and Informatics Program
of American University of Central Asia in Partial Fulfillment of the
Requirements for the Degree of
Bachelor of Arts

Author

Papakov Aidarbek

Certified by Thesis Supervisor

Dolmatova Polina

Accepted by

Sergey Sklyar

Head of Applied Mathematics and
Informatics Program, AUCA

May 2026

Bishkek, Kyrgyz Republic

ABSTRACT

Earthquake Early Warning (EEW) systems are intended to issue alerts before destructive seismic waves reach populated areas to minimize the damage caused by the seismic waves. One of the main challenges in such systems is estimating the final earthquake magnitude from only the first few seconds of seismic waveform observations. Traditional approaches, such as STA/LTA detection and Bayesian estimation frameworks, are limited in their ability to extract reliable magnitude estimates from such short observation windows.

This work presents an approach that uses both classical signal processing techniques along with deep learning to estimate earthquake magnitude based only on the first seconds of P-wave observations. The proposed architecture combines **Convolutional Neural Networks** (CNNs) for spatial feature extraction, a **Long Short-Term Memory** (LSTM) network for temporal modeling, followed by a **Multi-Layer Perceptron** (MLP), which serves as a regression head.

The proposed model is trained and evaluated on the INSTANCE dataset provided by the Italian National Institute of Geophysics and Volcanology (INGV), which contains approximately 1.2 million three-component seismic waveform data instances. To investigate the relationship between the model's performance and the length of the available time window, experiments are conducted with P-wave windows of 3, 5, 8, and 10 seconds. Ultimately, model performance is evaluated using traditional regression metrics: Mean Residual (Bias), Mean Absolute Error (MAE), the coefficient of determination (R^2), and Spearman's rank correlation coefficient.

The results indicate that the proposed CNN–LSTM–MLP architecture manages to successfully learn meaningful data representations from spectrograms and produces reliable estimates of magnitude from early P-wave observations. The best configuration achieves $MAE = 0.3152$ and $R^2 = 0.7478$ on a 10-second window. The results imply that spectrogram-based deep learning models are a promising direction for improving the current earthquake early warning systems.

TABLE OF CONTENTS

Abstract.....	2
Chapter 1. Introduction.....	6
1.1 Motivation.....	6
1.2 The Core Challenge of Early Magnitude Estimation.....	7
1.3 Limitations of Classical EEW Approaches.....	7
1.4 The Potential of Deep Learning in EEW.....	8
1.5 Proposed Approach.....	9
1.6 Scope and Limitations.....	9
1.7 Thesis Structure.....	10
Chapter 2. Background and Literature Review.....	11
2.1 Fundamentals of Seismology.....	11
2.2 Classical Earthquake Early Warning Systems.....	12
2.3 Data Sources for Seismic Machine Learning.....	13
2.4 Machine Learning Approaches to Seismic Monitoring.....	13
2.5 Limitations of Existing Approaches.....	15
2.6 Research Gap.....	15
2.7 Summary.....	15
Chapter 3. Mathematical Foundations.....	17
3.1 Overview.....	17
3.2 Signal Representation: From Waveform to Spectrogram.....	17
3.3 Convolutional Neural Networks.....	19
3.4 Recurrent Neural Networks and LSTMs.....	20
3.5 Loss Functions.....	21
3.6 Optimization.....	22
3.7 Evaluation Metrics.....	22
3.8 Summary.....	23
Chapter 4. Dataset and Preprocessing.....	24
4.1 Overview.....	24
4.2 Data Source.....	24

4.3 Preprocessing Pipeline.....	25
4.4 Dataset Splitting.....	26
4.5 Summary.....	27
Chapter 5. Model Architecture.....	28
5.1 Overview of the Model Pipeline.....	28
5.2 Input Representation.....	28
5.3 CNN Backbone.....	28
5.4 Feature Sequence Extraction.....	29
5.5 LSTM Module.....	29
5.6 Regression Head.....	30
5.7 Learned Feature Representations.....	31
5.8 Summary.....	32
Chapter 6. Experiments and Evaluation.....	33
6.1 Overview.....	33
6.2 Experimental Environment.....	33
6.3 Training Configuration.....	33
6.4 Experimental Design.....	35
6.5 Discussion.....	36
6.6 Visualization of Results.....	37
6.7 Summary.....	38
Chapter 7. Limitations.....	39
7.1 Overview.....	39
7.2 Data Imbalance.....	39
7.3 Single-Station Analysis.....	39
7.4 Dependence on P-wave Picking.....	39
7.5 Regional Variability.....	40
7.6 Summary.....	40
Chapter 8. Conclusion.....	41
8.1 Summary of the Work.....	41
8.2 Key Findings.....	41

8.3 Contributions.....	42
8.4 Future Work.....	42
8.5 Final Remarks.....	42
References.....	44

CHAPTER 1. INTRODUCTION

1.1 Motivation

Earthquakes remain one of the most dangerous natural disasters, posing a threat to both infrastructure and people. To this day, it is not an easy task to predict an earthquake due to the lack of explicit precautionary signals detectable by traditional monitoring systems. As a result, communities located near seismically active regions tend to experience catastrophic damage caused by strong ground motion. The consequences of large earthquakes on both the economy and society are therefore severe, including deaths and substantial damage to infrastructure [1].

To minimize the damage caused by earthquakes, specialized systems are developed and maintained. These systems are called Earthquake Early Warning (EEW) systems. They issue alerts as soon as an earthquake begins, but before the most destructive waves reach populated areas. The fundamental principle behind EEW is based on the difference in propagation speed between seismic waves. Two main types of seismic waves can be distinguished: P-waves and S-waves. The distinction between them can be summarized as follows: P-waves are faster, but significantly less destructive than S-waves [2].

This difference in velocities provides a short time window during which early warning systems can detect the initial P-waves and estimate the scale of the upcoming event before the arrival of destructive waves. In the context of EEW, even several seconds can have a substantial impact, as they might allow for activation of safety protocols such as stopping trains, shutting down industrial processes, opening fire station doors, and issuing alerts to the public [3].

Nevertheless, producing reliable warnings remains a challenge. EEW systems must process the propagating signal and decide whether the current event is notification-worthy in near real time, while operating on noisy sensor data.

It is also beneficial to obtain a reasonable estimate of the potential earthquake's magnitude in order to have more operational opportunities, however, with these possibilities, the difficulties also rise. The estimation is essential because alert thresholds are largely defined in terms of expected ground motion intensity, which correlates with earthquake magnitude. If the magnitude is underestimated, warnings may not be issued. On the contrary, overestimation leads to false alarms, ultimately resulting in public distrust of the warning system [4].

Traditional EEW methods are not primarily designed for magnitude estimation, and while research is being conducted in this direction, such methods largely rely on hand-crafted features extracted from the incoming signal, which require profound expertise in both seismology and the specific seismic patterns of the region studied. These limitations push the exploration of modern

machine learning approaches capable of learning to estimate magnitude from complex and noisy seismic sensor data, making the resulting models more region-agnostic [5].

1.2 The Core Challenge of Early Magnitude Estimation

To simplify, an earthquake's magnitude represents the total energy released during rupture along a fault. Determining the magnitude traditionally requires gathering a substantial portion of the seismic waveform, including both P-waves and S-waves recorded across multiple stations. In classical seismological analysis, magnitude computation is therefore performed after the event has ended [6].

In an EEW context, however, such enriched information is not yet available at the moment of decision-making. And the system must evaluate the final magnitude using only the earliest observations of the seismic signal, a short window beginning at the arrival of the P-wave and extending for several seconds.

Formally, consider a seismic waveform $x(t)$ recorded at a station, and let t_P denote the arrival time of the P-wave. An EEW system may only have access to the segment:

$$x(t), \quad t \in [t_P, t_P + W]$$

where W is the available observation window, often between 3 and 10 seconds. The task is to estimate the final earthquake magnitude M using only this limited segment of the signal.

This problem is challenging on multiple levels: the earliest portion of the waveform does not guarantee sufficient information for a reliable, precise magnitude decision. The early parts of the waveform may appear similar for both small and large events, making them difficult to distinguish using simple methods or heuristics. Several studies have discussed whether the initial observations contain enough information to make a definitive decision on magnitude. Some earlier hypotheses suggested that the first few seconds already contain distinguishable differences between small and large events. However, more recent research indicates that reliable magnitude estimation from very short windows remains an open challenge [7].

Therefore, an effective EEW system must extract implicit patterns from the evolving waveform that may indicate the scale of the ongoing rupture. This is precisely the task at which modern machine learning techniques excel.

1.3 Limitations of Classical EEW Approaches

Traditional EEW systems rely on deterministic signal processing algorithms and empirical relationships derived from seismological observations. The most widely used approach is the STA/LTA algorithm.

1.3.1 STA/LTA Detection

The Short-Term Average to Long-Term Average (STA/LTA) algorithm is one of the earliest methods for earthquake detection [8]. The method computes two moving averages of signal amplitude: a short-term average (STA) representing recent signal energy, and a long-term average (LTA) representing background noise. The detection statistic is defined as the ratio:

$$R(t) = \frac{\text{STA}(t)}{\text{LTA}(t)}$$

When this ratio exceeds a predefined threshold, the algorithm identifies the presence of a seismic event. While STA/LTA is effective for detecting sudden increases in signal energy, it cannot produce reliable estimates of earthquake magnitude, especially in the early stages of the waveform.

1.4 The Potential of Deep Learning in EEW

Recent advances in deep learning have significantly extended the capabilities of signal analysis systems due to their ability to learn feature representations from complex and highly correlated data, eliminating the need for manually engineered features.

In the context of seismology, deep learning has already demonstrated strong performance in tasks such as seismic phase detection, event classification, and waveform denoising [9, 10]. The key advantage of deep learning over traditional methods lies in its ability to approximate complex nonlinear mappings between inputs and outputs. Formally, a neural network can be viewed as a function:

$$f_{\theta} : \mathbb{R}^n \rightarrow \mathbb{R}$$

parameterized by a set of learnable weights θ . During training, these parameters are updated to minimize a loss function that measures the discrepancy between the network's predictions and the actual values.

When applied to seismic signals, neural networks can learn to recognize patterns in waveform data that correlate with earthquake magnitude. These patterns may involve subtle

combinations of frequency content, temporal evolution, and signal amplitude that are difficult to capture using conventional methods.

1.5 Proposed Approach

This work proposes a deep learning pipeline for earthquake magnitude estimation using spectrograms of early P-wave observations. The system analyzes three-component seismic waveforms and returns a single scalar value, which represents an estimate of earthquake magnitude.

The proposed model architecture is composed of Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) blocks, and Multi-Layer Perceptrons (MLPs), designed on the premise of the time–frequency domain provided by spectrograms. The processing pipeline can be summarized as follows:

- **Waveform preprocessing:** Three-component seismic signals are retrieved and normalized to a consistent scale.
- **Time–frequency transformation:** The Short-Time Fourier Transform (STFT) is applied to each waveform, resulting in three spectrograms representing seismic energy distribution across frequency and time.
- **CNN feature extraction:** Convolutional neural networks process the spectrograms to localize patterns in the time–frequency domain.
- **Temporal modeling with LSTM:** The resulting sequence of features extracted by the CNN is fed to an LSTM network to better model the temporal evolution of the signal.
- **Magnitude regression:** A multi-layer perceptron produces a scalar estimate of earthquake magnitude.

The motivation behind this CNN–LSTM design lies in the physical properties of seismic signals. Spectrograms represent the distribution of frequency content over time, reflecting the dynamics of the rupture. While CNNs excel at extracting local spatial patterns from such representations they struggle with temporal modeling which can be solved by LSTMs. By combining these two paradigms, the system aims to infer earthquake magnitude from features present in the earliest stages of the waveform.

1.6 Scope and Limitations

This work focuses specifically on single-station magnitude estimation using early P-wave observations. The study investigates whether deep learning models can reliably extract meaningful hidden representations from spectrograms and, consequently, estimate earthquake

magnitude using the first 3–10 seconds following P-wave arrival. The analysis is conducted using the publicly available INSTANCE seismic dataset [11].

However, several limitations must be noted:

- **Prospective evaluation:** The experimental results are intended as a proof-of-concept investigation rather than a fully deployed operational EEW system.
- **Single-station analysis:** The proposed model uses observations from a single station, whereas modern EEW systems usually rely on observations from multiple stations.
- **Data imbalance:** As earthquake magnitude distribution follows the Gutenberg–Richter law [12], large earthquakes are significantly less frequent than small ones, naturally creating an imbalance that may introduce bias during training and evaluation.
- **Regional variability:** Seismic waveforms depend on local geological conditions. Models trained on data from one region must be additionally fine-tuned or completely retrained for optimal performance in another region.

Despite these limitations, the proposed approach provides a framework for exploring the use of deep learning combined with spectrograms for early earthquake magnitude estimation.

1.7 Thesis Structure

The remainder of this thesis is organized as follows.

Chapter 2 provides background information on seismological principles and reviews existing earthquake early warning systems, including both classical and machine learning approaches.

Chapter 3 introduces the mathematical foundations of the proposed methodology, formally defining the signal processing techniques, neural network architectures, optimization methods, and evaluation metrics used in the study.

Chapter 4 describes the dataset and its preparation, including preprocessing and filtration steps.

Chapter 5 presents the architecture of the proposed CNN–LSTM model, including detailed descriptions of the convolutional backbone, recurrent module, and regression head.

Chapter 6 outlines the experimental design and evaluation methodology used to assess model performance, and presents both the results and a discussion of their implications.

Chapter 7 discusses the limitations of the proposed approach.

Chapter 8 summarizes the key findings of the thesis and suggests directions for further development of the proposed method.

CHAPTER 2. BACKGROUND AND LITERATURE REVIEW

2.1 Fundamentals of Seismology

Before describing the proposed method in detail, it is necessary to introduce several terms related to seismology.

2.1.1 Earthquake Generation

Earthquakes result from a sudden release of accumulated strain energy in the Earth's crust. Tectonic stresses gradually deform rocks along geological faults until the stress exceeds the frictional resistance preventing slip. At that point, rapid rupture propagates along the fault plane, releasing the stored energy in the form of seismic waves [13].

The point inside the Earth where the rupture initiates is called the hypocenter (or focus), whereas the projection of this point onto the Earth's surface is called the epicenter. Depending on the size of the earthquake, the rupture process may propagate along the fault plane over several seconds or longer.

2.1.2 Seismic Wave Types

As noted above, there are several distinguishable types of seismic waves. The main two are P-waves (Primary or Pressure waves) and S-waves (Secondary or Shear waves).

P-waves are faster and therefore the first to reach the Earth's surface, however, they are less destructive than S-waves. The velocity of P-waves typically ranges from approximately 5 to 8 km/s in the Earth's crust [2].

S-waves travel notably slower than P-waves but are typically more destructive. Their velocity ranges from approximately 3 to 5 km/s in crustal rocks. The difference in velocities forms the basis for most early warning systems deployed around the world, as the time window between P-wave and S-wave arrival allows warnings to be issued before the destructive waves reach vulnerable areas.

2.1.3 Earthquake Magnitude and Intensity

The scale of an earthquake can be measured in several ways. Historically, the Richter magnitude scale was used to estimate earthquake size based on the logarithm of the maximum amplitude recorded by a seismograph. However, this scale saturates for large earthquakes and is no longer widely used for scientific analysis [14].

Modern seismology relies on the moment magnitude scale, M_w , derived from the seismic moment:

$$M_0 = \mu AD$$

where μ is the shear modulus of the rock, A is the rupture area, and D is the average slip along the fault. The moment magnitude is then defined as:

$$M_w = \frac{2}{3} \log_{10}(M_0) - 6.07$$

This provides a consistent measure of earthquake scale across a wide range of magnitudes [14].

It is also important to distinguish magnitude from intensity, which measures how severely the ground shakes at a particular location. The Modified Mercalli Intensity (MMI) scale ranges from I (not felt) to XII (complete destruction). Because intensity is highly dependent on local soil conditions and other site-specific factors, it is not region-agnostic, consequently, EEW systems typically estimate earthquake magnitude and location rather than intensity.

2.2 Classical Earthquake Early Warning Systems

Traditionally, EEW systems have relied on signal processing methods and empirically derived relationships defined by domain experts. The most common approach is energy-based detection algorithms.

2.2.1 STA/LTA Detection

The Short-Term Average / Long-Term Average (STA/LTA) algorithm is one of the simplest yet most widely used methods for seismic event detection. The algorithm calculates the ratio between two moving averages of signal amplitude. The short-term average represents recent signal energy, while the long-term average represents background noise. The detection statistic is then defined as:

$$R(t) = \frac{\text{STA}(t)}{\text{LTA}(t)}$$

When this ratio exceeds a predefined threshold set by domain experts, the system signals a seismic event. The STA/LTA algorithm is incredibly light-weight and due to this it operates in real time. However, it primarily detects abrupt increases in signal amplitude and provides little or

no information about the physical properties of the earthquake, such as magnitude. It is also sensitive to noise produced by infrastructural sources such as vehicles, construction work, and similar activities, consequently, it may produce false positives in environments with frequent background vibrations.

2.3 Data Sources for Seismic Machine Learning

2.3.1 Ground-Based Seismic Sensors

The most common seismic monitoring instruments are seismometers and accelerometers installed at ground stations. Such instruments measure ground motion along three orthogonal axes, producing three-component waveform recordings. Ground-based sensors are the primary data source in this work due to their high-quality recordings, accessibility and their ability to detect subtle ground motions associated with small earthquakes, which constitute the largest portion of all seismic events.

2.3.2 Global Navigation Satellite Systems

Although not used in this work, Global Navigation Satellite Systems (GNSS) deserve mention. GNSS receivers measure ground displacement directly by tracking changes in satellite signal timing, and can complement seismic data by providing accurate estimates of large ground displacements associated with major earthquakes. However, GNSS measurements are significantly noisier than seismic recordings, making them effective only for very large earthquakes. Studies have shown that GNSS-only systems suffer from an inability to detect small and medium earthquakes ($M < 6.0$), as the received signals often contain insufficient information [15]. Consequently, most EEW systems rely primarily on seismic sensors for early detection, with GNSS measurements serving as complementary data for very large events.

2.4 Machine Learning Approaches to Seismic Monitoring

With the rapid development of deep learning over the past decade, researchers have increasingly explored machine learning applications for seismic signal analysis. Neural networks in particular have demonstrated strong performance in tasks such as seismic phase picking, earthquake detection, magnitude estimation, and ground motion prediction.

2.4.1 Machine Learning Advances and Achievements

Among the notable advances, PhaseNet stands out as a deep neural network designed for automatic detection of P-wave and S-wave arrivals [9]. Based on a convolutional encoder–decoder structure similar to the U-Net architecture, PhaseNet processes raw seismic

waveforms and learns representations directly from labeled training data, producing probability estimates for phase arrivals at each time step. PhaseNet has been shown to outperform traditional STA/LTA-based methods in both detection accuracy and robustness to noise.

Another relevant system is CRED (**CNN–RNN Earthquake Detector**), which represents a deep learning approach for earthquake detection using spectrograms of three-component seismic observations [16]. The network is composed of convolutional and recurrent neural networks, exploiting spectrograms as a rich spatio-temporal representation of seismic waves. CRED is trained to solve a binary classification task, outputting 1 for earthquake presence and 0 otherwise. This work shares architectural similarities with the proposed approach, but addresses a fundamentally simpler task than scalar magnitude estimation.

M-LARGE is another approach for estimating earthquake magnitude [17]. It utilizes GNSS sensor data and, as concluded by the authors themselves, is practically ineffective for low-to-medium-magnitude earthquakes. Nevertheless, the proposed model performed exceptionally well on synthetically generated high-magnitude data, indicating that deep learning models may excel at capturing complex patterns in seismic data regardless of the input modality.

Other directions of research focus on directly estimating earthquake parameters from raw waveform data. Some researchers have concatenated three-component waveforms into a single matrix as input to CNNs, while others have used single waveforms as input to RNN-based architectures, aiming to let the models learn relevant features directly from the raw data.

Motivated by the success of attention-based architectures in Computer Vision and Natural Language Processing, the Transformer Earthquake Alerting Model (TEAM) combines convolutional layers with transformer-based attention mechanisms to estimate probability distributions of peak ground acceleration (PGA) [18]. The architectural choices allow capturing long-range temporal dependencies within seismic waveforms. Although such models demonstrate promising performance, their computational complexity can pose challenges for real-time EEW deployment.

Several models have been proposed specifically for magnitude estimation. For example, MagNet estimates earthquake magnitude from seismic waveforms and demonstrates reasonable performance for small to moderate events, although the model performs best in the lower-to-mid range of magnitudes due to the distribution of the training data being skewed toward smaller events [19].

Deep learning has also found application in signal denoising. Zhu and Mousavi adapted a U-Net-like architecture to remove noise from incoming seismic signals, with the extrinsic evaluation metric being the improved STA/LTA performance on denoised signals compared to original waveforms [20].

2.5 Limitations of Existing Approaches

Despite significant progress in machine learning-based seismic monitoring, several limitations persist.

- **Dependence on full waveforms:** A significant portion of studies utilize full waveforms, including both P-waves and S-waves. While this improves model metrics, it reduces the relevance of such systems for real-time early warning, where only the earliest seconds of data are operationally valuable.
- **Data imbalance:** The naturally inherited imbalance in magnitude data implies that datasets used for model training and experimentation are dominated by low-magnitude events, which makes it difficult for models to learn reliable representations of large earthquakes.
- **Underutilization of signal processing:** Most proposed architectures rely solely on raw inputs rather than utilizing established signal processing transformations such as frequency-domain spectrograms.

2.6 Research Gap

The literature reviewed above highlights several opportunities for further research.

First, relatively few studies focus explicitly on estimating earthquake magnitude using only the earliest seconds of P-wave data, most approaches rely on longer waveform segments that include S-wave observations.

Second, many architectures address either spatial representations, temporal modeling, or frequency-domain features individually, only a small number of approaches combine all three simultaneously. Systems that do combine all three typically address the simpler task of event classification rather than estimation of a continuous scalar value.

Finally, the scarcity of high-magnitude training data remains a major challenge. As a consequence, most research is conducted using highly skewed data (M2.0-3.0). Techniques such as synthetic data generation and targeted sampling strategies may help mitigate this limitation.

2.7 Summary

This chapter reviewed the physical principles underlying earthquake generation and seismic wave propagation, the traditional methods used in earthquake early warning systems, and advances made within the deep learning field. Classical EEW algorithms rely on signal processing techniques and manually tuned parameters that require substantial domain expertise

in both seismology and regional characteristics. Such methods struggle to estimate earthquake magnitude accurately within the first few seconds of an event.

Recent deep learning advances have introduced new approaches for seismic signal analysis. However, challenges remain, including dataset imbalance, reliance on full waveform data, and limited exploration of architectures combining spatial and temporal feature extraction. These limitations directly motivate the methodology proposed in this thesis, which integrates spectrograms, convolutional neural networks, and recurrent neural networks to estimate earthquake magnitude from early P-wave observations.

CHAPTER 3. MATHEMATICAL FOUNDATIONS

3.1 Overview

The proposed method consists of several components, each motivated by advances in seismology as well as adjacent fields such as image processing and natural language processing. The pipeline can be illustrated as follows:

Three-component waveform $\rightarrow$ spectrogram representations $\rightarrow$ CNN $\rightarrow$ LSTM $\rightarrow$ MLP.

The spectrogram provides a time–frequency representation of the seismic signal. Convolutional neural networks serve as spatial feature extractors, while recurrent neural networks capture temporal dependencies. Finally, a multi-layer perceptron predicts the earthquake magnitude from the extracted features. This chapter introduces the mathematical foundations underlying these components.

3.2 Signal Representation: From Waveform to Spectrogram

Seismic sensors record signals in the time domain. For a single seismic channel, the recorded waveform can be represented as a function $x(t)$, where t denotes time and $x(t)$ represents ground displacement, velocity, or acceleration depending on the instrument type. Although time-domain waveforms contain all information about the seismic signal, many characteristics of earthquakes are better understood in the frequency domain due to the noise introduced by infrastructural sources.

3.2.1 The Short-Time Fourier Transform

The classical Fourier transform converts a time-domain signal into its frequency representation:

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$$

where ω is angular frequency and $j = \sqrt{-1}$. However, the Fourier transform assumes that the signal is stationary over the entire observation period. Seismic signals are non-stationary, as their properties change notably during the progression of the earthquake.

To capture these changes, the Short-Time Fourier Transform (STFT) is used, applying the Fourier transform to short overlapping windows of the signal:

$$X(\tau, \omega) = \int_{-\infty}^{\infty} x(t)w(t - \tau)e^{-j\omega t}dt$$

where $x(t)$ is the input signal, $w(t - \tau)$ is a window function centered at time τ , and ω represents frequency. In practice, seismic signals are discrete sequences with sampling interval T_s . The discrete STFT is therefore used:

$$X[m, k] = \sum_{n=0}^{N-1} x[n]w[n - mH]e^{-j\frac{2\pi kn}{N}}$$

where m is the frame index, k is the frequency bin index, H is the hop length (stride between windows), and N is the Fast Fourier Transform (FFT) size. This transform produces a complex-valued matrix describing signal energy at different time and frequency locations.

3.2.2 Time–Frequency Resolution Tradeoff

A fundamental limitation of time–frequency analysis arises as time resolution Δt and frequency resolution Δf satisfy the following inequality:

$$\Delta t \cdot \Delta f \geq \frac{1}{4\pi}$$

It is then implied that increasing time resolution necessarily decreases frequency resolution, and vice versa. In the context of seismic signals sampled at 100 Hz, the choice of window length must balance these competing requirements. Because EEW systems rely on short observation windows (typically 3–8 seconds), STFT parameters must be chosen carefully to preserve relevant signal structure.

3.2.3 Window Function

A Hann window function is used in this work, which is expressed as:

$$w[n] = 0.5 \left(1 - \cos \left(\frac{2\pi n}{N-1} \right) \right), \quad 0 \leq n \leq N-1$$

where N is the FFT size. The choice of this specific window function helps reduce spectral leakage at window boundaries by tapering the values to zero on both ends, in contrast to a rectangular window.

3.2.4 Magnitude Spectrogram

The STFT produces a complex-valued matrix. To use the values produced by the STFT, we convert them to the real domain by taking the magnitude of these coefficients, which yields a data matrix suitable for machine learning:

$$S[m, k] = |X[m, k]|$$

The resulting magnitude spectrogram provides a numerically stable representation that we can pass to our architecture.

3.3 Convolutional Neural Networks

Once seismic waveforms have been converted into spectrograms, the resulting representations can be viewed as two-dimensional matrices with axes corresponding to time and frequency. Convolutional Neural Networks (CNNs) by their nature, are suited to such spatial data [21].

3.3.1 Convolution Operation

Consider an input feature map $X \in \mathbb{R}^{H \times W}$. A convolution kernel $K \in \mathbb{R}^{k_h \times k_w}$ slides across the input and computes weighted sums:

$$(X * K)[i, j] = \sum_m \sum_n X[i + m, j + n] K[m, n]$$

The output of the convolution operation is a feature map that highlights patterns detected by the kernel. In CNNs with multiple channels, each output channel is computed as:

$$Z_c^{(l)} = b_c + \sum_{k=1}^{C_{\text{in}}} K_{c,k}^{(l)} * X_k^{(l-1)}$$

3.3.2 Activation Functions

After convolution, nonlinear activation functions introduce nonlinearity into the network. While the sigmoid function was used in early machine learning, it is now rarely used due to the vanishing gradient problem. The standard activation function in modern architectures is the Rectified Linear Unit (ReLU):

$$f(z) = \max(0, z)$$

ReLU avoids the saturation effects present in sigmoid or hyperbolic tangent functions, reducing the risk of vanishing gradients during training.

3.3.3 Pooling Layers

Pooling layers reduce the spatial dimensions of feature maps, decreasing computational cost. Max pooling selects the maximum value in each pooling region, while average pooling computes the mean. In the proposed architecture, max pooling is chosen over average pooling between convolutional blocks due to empirically observed superior performance. At the end, an adaptive average pooling layer ensures a fixed output shape regardless of minor input length variations.

3.3.4 Batch Normalization

Batch normalization has become indispensable in deep networks [22]. It normalizes activations across a mini-batch:

$$\hat{x}_i = \frac{x_i - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}$$

where μ_B is the batch mean and σ_B^2 is the batch variance. The normalized activations are then scaled and shifted using learnable parameters, stabilizing gradient propagation and improving model convergence.

3.3.5 Dropout

To address overfitting, a phenomenon when a model performs significantly better on training data than on validation data, dropout is used as a regularization technique [23]. During training, each neuron is randomly deactivated with probability p and to employ this at inference time, the outputs of all neurons are scaled by $(1 - p)$ to compensate for the neurons dropped

during training. Dropout has been shown to improve the generalization capabilities of deep learning models.

3.4 Recurrent Neural Networks and LSTMs

While CNNs are well suited to processing the spatial structure of spectrograms, the processed signals still retain a time-dependent component. Modeling temporal dynamics requires sequential neural architectures.

3.4.1 Vanilla Recurrent Neural Networks

A recurrent neural network processes sequences by maintaining a hidden state. For an input sequence $x_1, x_2, \dots, x_T$, the hidden state update is:

$$h_t = \tanh(W_h h_{t-1} + W_x x_t + b)$$

The main limitation of vanilla RNNs is the vanishing gradient problem: during backpropagation, gradients propagate through many sequential steps and often decrease asymptotically, rendering learning ineffective over long sequences.

3.4.2 Long Short-Term Memory Networks

To address the issue of vanishing gradients, Long Short-Term Memory (LSTM) networks were introduced [24]. An LSTM cell maintains an internal state c_t that stores long-term information through several gating mechanisms:

$$\begin{aligned} f_t &= \sigma(W_f[h_{t-1}, x_t] + b_f) \\ i_t &= \sigma(W_i[h_{t-1}, x_t] + b_i) \\ o_t &= \sigma(W_o[h_{t-1}, x_t] + b_o) \end{aligned}$$

where f_t is the forget gate, responsible for choosing what information from c_{t-1} to omit, i_t is the input gate, responsible for filtering the data arriving at time step t , and o_t is the output gate, whose main task is to control what information flows to the next cell. Together, these mechanisms allow the network to selectively retain or discard information over long time intervals.

3.4.3 CNN–LSTM–MLP Integration

In the proposed architecture, the CNN serves as a backbone that processes spectrograms and produces feature vectors which are passed to the LSTM, it then models the temporal

evolution. This combination enables the model to capture both spatial patterns and temporal dependencies. The resulting feature vector is then passed to a multi-layer perceptron that ultimately predicts the scalar magnitude value.

3.5 Loss Functions

The objective of the model is to estimate earthquake magnitude M . The predicted magnitude is denoted $\hat{M}$. A loss function (also called a cost function) measures the discrepancy between the actual value (ground truth) and the model's output (predicted value), and is used to update the weights θ via gradient descent. The lower the loss, the more accurate the predictions. The loss function used in this work is the Huber loss:

$$L_{\delta}(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2 & \text{if } |y - \hat{y}| \leq \delta, \\ \delta|y - \hat{y}| - \frac{1}{2}\delta^2 & \text{otherwise.} \end{cases}$$

The value of δ in this work is set to 0.3, which was determined empirically to perform best after several experimental trials.

3.6 Optimization

Neural network parameters are optimized using gradient-based learning. The basic gradient descent update rule is:

$$\theta \leftarrow \theta - \eta \nabla_{\theta} L$$

where θ represents model parameters and η is the learning rate. Modern deep learning practitioners use variations of gradient descent. The most common ones are RMSProp, Adam, and AdamW [25]. In this work, the AdamW optimizer is utilized, which maintains adaptive learning rates using estimates of the first and second moments of the gradient.

3.7 Evaluation Metrics

To evaluate model performance on the scalar regression task, the following metrics are used.

Mean Residual:

$$\text{Bias} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)$$

Mean Absolute Error:

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$$

Spearman's correlation:

$$\rho = \frac{\sum_{i=1}^N (r_i - \bar{r})(s_i - \bar{s})}{\sqrt{\sum_{i=1}^N (r_i - \bar{r})^2 \cdot \sum_{i=1}^N (s_i - \bar{s})^2}}$$

Coefficient of Determination:

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

Bias measures the tendency of the model to overestimate or underestimate the magnitude, with values near zero implying that the model does neither. MAE is the primary metric, due to its interpretability, and is also used to select the best model configuration. Spearman's correlation serves as a measure of agreement between the ground truth and the model's predictions. Finally, the coefficient of determination is tracked to evaluate how much of the variance in the target the model accounts for.

3.8 Summary

This chapter presented the mathematical principles underlying the proposed magnitude estimation system. The Short-Time Fourier Transform provides a time–frequency representation of seismic signals. Convolutional neural networks extract spatial patterns from spectrograms, while recurrent networks model temporal dependencies. A regression MLP then maps the learned features to a scalar magnitude estimate.

CHAPTER 4. DATASET AND PREPROCESSING

4.1 Overview

Although deep learning excels at processing complex, high-dimensional data, it still requires large amounts of labeled examples to perform at a sufficient level. While there is a considerable amount of earthquake data available worldwide, there is significantly less data, and even fewer ready-to-use datasets, for machine learning experimentation. In the context of this study, the training dataset must contain both three-component waveform recordings and the associated ground truth magnitude value for each event. To test the proposed approach, the INSTANCE dataset was used [11]. The main reasons for choosing this dataset are its verbose metadata, which contains a detailed description for each column and allows the data to be filtered for our specific task, and its size, with a large amount of data distributed across a geographically dense region, which ensures that all the data come from a single distribution.

4.2 Data Source

4.2.1 The INSTANCE Dataset

The INSTANCE dataset is, as stated by its authors, intended for machine learning experiments. It was developed by the Italian National Institute of Geophysics and Volcanology (INGV) [11]. The dataset contains approximately 1.2 million waveform recordings, each with the following properties:

- three-component recordings (vertical, north–south, east–west).
- waveform duration of 120 seconds.
- sampling rate of 100 Hz.
- magnitudes range from approximately 0.0 to 6.5.
- phase picking type: manual and automated.

According to the authors, the dataset underwent several preprocessing steps: removal of traces containing data gaps, trimming of waveform segments to consistent start times, resampling to a uniform sampling frequency, and removal of mean offsets and linear trends.

4.2.2 Dataset Filtration

Several filters were applied to the original dataset to meet the requirements of this study. First, only observations measured in m/s were retained, ensuring consistent units of measurement. Second, observations with automatically picked phases were removed. Although automatic phase picking does not necessarily imply inconsistent or incorrect results, for the sake

of dataset reliability we adopted the good practice of using only data whose labels have been verified by human experts. This reduces the potential bias introduced by unverified automated labels.

As stated above, the original dataset contains around 1.2 million observations. The distribution of magnitudes is heavily skewed toward the lower end, which must be addressed. To address this, we use an undersampling technique. Undersampling was chosen over upsampling or data augmentation because the latter methods require deep knowledge of seismology to be applied correctly. To avoid making domain-specific mistakes, those approaches were omitted. The full dataset is then sampled in the following proportions:

- 50,000 earthquakes with magnitudes in the range M1.0–M3.0.
- 50,000 earthquakes with magnitudes in the range M3.0–M5.0.
- all available earthquakes with magnitude greater than M5.0.

The primary goal of this work is to estimate the relationship between early P-wave observations and source magnitude. Training with an extreme cluster of low-magnitude events would not suit this objective.

4.3 Preprocessing Pipeline

Although the authors of the INSTANCE dataset applied initial preprocessing, further steps are required before the data can be used to train the proposed model. The following pipeline transforms the waveform into a standardized representation suitable for model training.

4.3.1 P-wave Detection and Window Extraction

EEW systems rely on the initial P-wave arrival. Once the P-wave arrival time t_P is known, a short window following the arrival is extracted for processing. For a waveform $x(t)$, the window is defined as:

$$x(t), \quad t \in [t_P, t_P + W]$$

where W is the window length. In this study, several window lengths are considered:

$$W \in \{3, 5, 8, 10\} \text{ seconds}$$

These multiple window lengths are selected to study the dependence of the learned relationship on the amount of data provided.

4.3.2 Signal Normalization

Waveforms are normalized before being passed through the STFT to ensure a consistent input scale for the model. The following normalization is applied:

$$\hat{x}(t) = \frac{\tilde{x}(t) - \mu_{\tilde{x}}}{\max_t |\tilde{x}(t) - \mu_{\tilde{x}}| + \epsilon}$$

where $\tilde{x}(t)$ is the linearly detrended signal, $\mu_{\tilde{x}}$ is its channel-wise mean, and $\epsilon = 10^{-8}$ prevents division by zero.

4.3.3 Spectrogram Computation

After window extraction and normalization, the waveform is converted into a time–frequency representation using the Short-Time Fourier Transform described in the previous chapter. For three-component seismic signals, the spectrogram is computed for each channel separately.

4.3.4 Input Tensor Construction

The spectrograms from the three seismic channels are stacked to form a three-dimensional tensor:

$$X \in \mathbb{R}^{3 \times F \times T}$$

where 3 is the number of seismic channels, F is the number of frequency bins, and T is the number of time frames. This tensor is passed to the CNN as the model input.

4.4 Dataset Splitting

To evaluate model performance reliably, the dataset is divided into separate subsets for training, validation, and testing:

- training set: 70% of the data.
- validation set: 20% of the data.
- test set: 10% of the data.

The validation set is used to evaluate the model's performance during training, the test set is used after training is completed, to ultimately verify the model's performance. All subsequent graphs and metrics reported in this thesis are computed on the test set. The split is performed using stratified sampling based on magnitude bins, ensuring that earthquakes of different magnitudes are represented proportionally in each subset.

4.5 Summary

This chapter described the data and preprocessing steps used in the proposed earthquake magnitude estimation system. The INSTANCE dataset provides a sufficient amount of data for stable model training. The preprocessing pipeline converts raw seismic waveforms into spectrograms suitable for model input, with the key steps including window extraction, normalization, and time–frequency transformation.

CHAPTER 5. MODEL ARCHITECTURE

5.1 Overview of the Model Pipeline

The goal of the proposed method is to estimate earthquake magnitude using only the first few seconds of seismic waveforms after the P-wave arrival. The overall architecture combines a CNN for spatial feature extraction, an LSTM module for modeling temporal dynamics, and a stack of three linear projections serving as a regression head.

Formally, the model can be expressed as:

$$\hat{M} = f_{\theta}(X)$$

where X is the raw three-component seismic waveform, θ is the set of learnable parameters, and $\hat{M}$ is the predicted earthquake magnitude. The architecture is built on two primary premises: spectral structure (frequencies vary distinctively during an earthquake) and temporal evolution (signal characteristics evolve critically during the first seconds after P-wave arrival).

5.2 Input Representation

The input to the neural network is a raw three-component seismic waveform tensor of shape (Batch, 3, 1000). The STFT is configured with $n_{fft} = 128$, $\text{win_length} = 128$, $\text{hop_length} = 38$, and a Hann window function. The resulting spectrogram tensor has the shape:

$$S \in \mathbb{R}^{3 \times F \times T}$$

Given the STFT configuration and the input length, the parameters resolve to $F = 65$ frequency bins and $T = 24$ time frames. This output tensor is treated similarly to a three-channel RGB image, allowing the CNN backbone to extract meaningful spatial patterns.

5.3 CNN Backbone

5.3.1 Convolutional Block Structure

The CNN backbone consists of a sequence of convolutional blocks designed to convert the 65×24 spectrogram into a dense feature representation. Each of the three blocks follows a standard deep learning pattern: $\text{Conv2D} \rightarrow \text{BatchNorm2D} \rightarrow \text{ReLU} \rightarrow \text{MaxPool2D}$. At the end

of the convolutional sequence, an AdaptiveAvgPool2D layer is used to guarantee a fixed spatial output shape of 1×16 , ensuring that downstream components receive a compatible data format.

5.3.2 Layer Configuration

The exact configuration of the convolutional layers is structured to progressively increase channel depth while reducing spatial dimensions:

Layer	Channels	Kernel Size	Stride	Padding	Pooling
Conv1	16	5×3	1	2×1	Max (2×1)
Conv2	32	3×3	1	1×1	Max (2×2)
Conv3	64	3×3	1	1×1	AdaptiveAvg (1×16)

Table 5.1. Configuration of the convolutional backbone.

5.4 Feature Sequence Extraction

After passing through the CNN backbone and the adaptive pooling layer, the resulting feature map has the shape $Z \in \mathbb{R}^{C' \times F' \times T'}$, with $C' = 64$, $F' = 1$, and $T' = 16$. The frequency dimension is squeezed out, leaving a tensor of shape (Batch, 64, 16). To feed this representation into the recurrent network, the tensor is permuted to swap the channel and time dimensions, yielding a sequence of vectors:

$$Z' \in \mathbb{R}^{T' \times D}$$

where $T' = 16$ time steps and $D = 64$ features. Each time step represents the condensed spectral feature representation extracted from a specific segment of the input waveform.

5.5 LSTM Module

5.5.1 Motivation

While CNNs capture local time–frequency patterns in spectrograms, they cannot explicitly model the longer-term temporal evolution of the signal. Earthquake magnitude is strongly related to how seismic energy evolves over time, for example through sustained increases in energy during the rupture phase.

5.5.2 LSTM Architecture

The model utilizes a single-layer LSTM network to process the sequence $Z' = (z_1, z_2, \dots, z_{16})$. The LSTM has a hidden size of $H_{\text{lstm}} = 64$ and outputs a sequence of 16 hidden states, one for each time step.

5.6 Regression Head

The resulting vector is then passed through a MLP to output the predicted magnitude. The regression head architecture is as follows:

Layer	Type	Configuration
1	Linear	$64 \rightarrow 128$
2	LayerNorm	128
3	Activation	ReLU
4	Dropout	$p = 0.4$
5	Linear	$128 \rightarrow 64$
6	LayerNorm	64
7	Activation	ReLU
8	Dropout	$p = 0.4$
9	Linear	$64 \rightarrow 1$

Table 5.2. Architecture of the regression head.

The final output layer produces the scalar magnitude estimate $\hat{M} \in \mathbb{R}$. No activation function is applied at the output node, as magnitude prediction is treated as a continuous regression task. The 40% dropout rate helps prevent overfitting on the dense layers.

5.7 Learned Feature Representations

For a long time, deep learning models have been criticized as a "black box" that one just puts some numbers in and gets some numbers out. The argument for validity of this judgement can be made to some extent, but within the past decade, researchers and practitioners have been trying different methods to see what deep learning models actually learn during the training process. The approach we have routed is the 2D projection of internal embedding space via the help of Principal Component Analysis (PCA).

The proposed model can be logically divided into two parts: a CNN–LSTM backbone, which serves as a feature extractor, and an MLP regression head, which is responsible for the final decision based on the embeddings produced by the backbone. To verify that the backbone extracts meaningful vector representations, these embeddings are analyzed.

Given a centered dataset of pooled LSTM output vectors X , PCA computes the covariance matrix C :

$$C = \frac{1}{n-1} X^T X$$

By performing eigendecomposition on this covariance matrix, eigenvectors v_i and eigenvalues λ_i are obtained:

$$Cv_i = \lambda_i v_i$$

Projecting the data onto the top two eigenvectors reduces the high-dimensional latent space while preserving meaningful distance relationships.

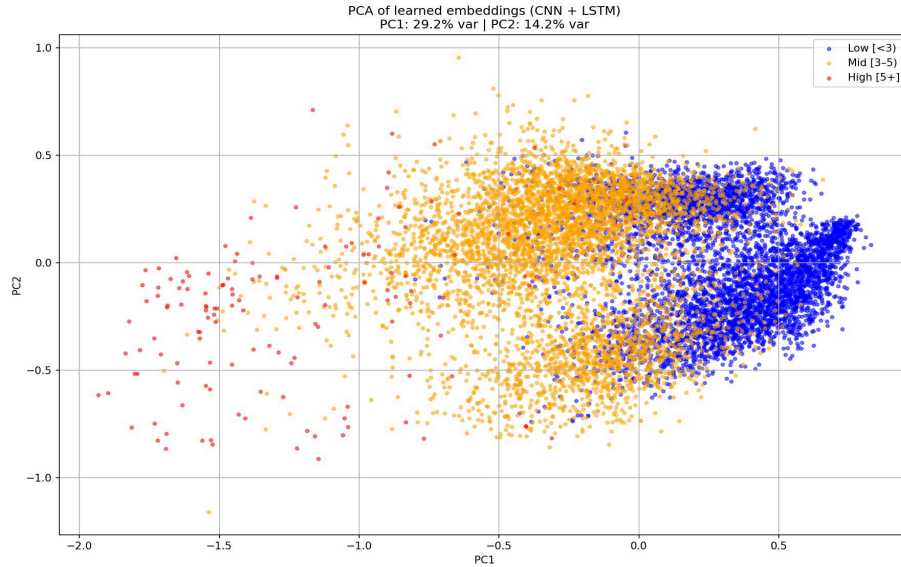


Figure 5.1. Two-dimensional PCA projection of the 64-dimensional pooled LSTM embeddings, colored by magnitude bucket: low ($M < 3$), mid ($M \in [3, 5]$), and high ($M \geq 5$). The first principal component accounts for 29.2% of the variance and the second for 14.2%.

The projection reveals the structure of the model's learned representations and confirms that the feature-extractor backbone successfully maps raw spectrograms into meaningful vector representations before the final regression step performed by the MLP. A clear magnitude gradient is visible along the first principal component: low-magnitude events (blue) concentrate on the right side of the projection, mid-magnitude events (orange) occupy the center, and high-magnitude events (red) lie predominantly on the left. The fact that this separation emerges from an unsupervised projection of the embeddings, even though PCA is unaware of magnitude

labels, suggests that the backbone has learned to map the spectrograms into a latent space that MLP can operate more easily on.

5.8 Summary

This chapter detailed the architecture of the proposed CNN–LSTM–MLP model for earthquake magnitude estimation. The model computes spectrograms on-device before processing them through a three-block convolutional backbone. Temporal dynamics are captured via a unidirectional LSTM. Finally, a regression MLP translates the condensed representation into a single magnitude estimate.

CHAPTER 6. EXPERIMENTS AND EVALUATION

6.1 Overview

This chapter describes the procedure for evaluating the proposed method, including the experimental setup, hyperparameters, results, and a discussion of their implications. The central question addressed is: can early observations of P-waves provide sufficient information for magnitude estimation?

6.2 Experimental Environment

6.2.1 Software Framework

All experiments are conducted using the Python programming language due to its rich infrastructure for machine learning experimentation. More specifically, PyTorch provides the core deep learning environment. Other libraries include NumPy for numerical operations, Pandas for data handling, Matplotlib for visualization, and Scikit-learn for evaluation metrics.

6.2.2 Hardware Configuration

Deep learning models perform a substantial amount of matrix multiplication operations, which modern GPUs handle well via CUDA kernels. The following configuration was used for all experiments:

Component	Specification
CPU	Intel Core i5-13400F
GPU	NVIDIA GeForce RTX 3070
RAM	16 GB

Table 6.1. Hardware configuration used for all experiments.

This configuration allows a meaningfully large batch size during training, which positively affects training stability and model convergence.

6.3 Training Configuration

6.3.1 Hyperparameter Tuning

As stated previously, a trained machine learning model can be expressed as a function of input x with parameters θ . However, training additionally requires a set of hyperparameters, parameters that are not learned via gradient descent, but set manually.

To find the right set of hyperparameters, two main approaches exist: Grid Search, which exhaustively tests all combinations in a predefined grid of values, and Random Search, which samples values from allowed ranges. Bergstra and Bengio demonstrated that random search performs better than grid search in practice [26]. Therefore, random search is used for hyperparameter tuning in this work.

Specifically, the hyperparameters are sampled from the following distributions. The learning rate and weight decay are drawn from log-uniform distributions in the ranges $[10^{-3.3}, 10^{-2.3}]$ and $[10^{-3.0}, 10^{-2.0}]$, respectively. Sampling from a log-uniform distribution is appropriate because both hyperparameters have a logarithmic-scale effect on the training process. The batch size is chosen from a discrete uniform distribution over the set $\{128, 256\}$. The δ parameter of the Huber loss is initially sampled from a continuous uniform distribution between 0.15 and 0.6. Nevertheless, subsequent analysis revealed that all of the best-performing experiments had $\delta \approx 0.3$, so this value was fixed. Early stopping is also used: training is halted if no improvement is observed for 150 consecutive epochs.

6.3.2 Optimization Algorithm

Model parameters are optimized using the AdamW optimizer [25]. The formal update equations are:

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t \\ v_t &= \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \\ \hat{m}_t &= \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t} \\ \theta_t &= \theta_{t-1} - \eta \left(\frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} + \lambda \theta_{t-1} \right) \end{aligned}$$

where η is the learning rate and λ is the weight decay coefficient, both of which are randomly sampled at the beginning of the training process as stated previously. m_t represents an exponential moving average of the gradients, while $\hat{m}_t$ is the corrected by β_1 . By analogy, v_t is an exponential moving average of squared gradients which is later adjusted via β_2 to become $\hat{v}_t$.

The remaining hyperparameters are fixed:

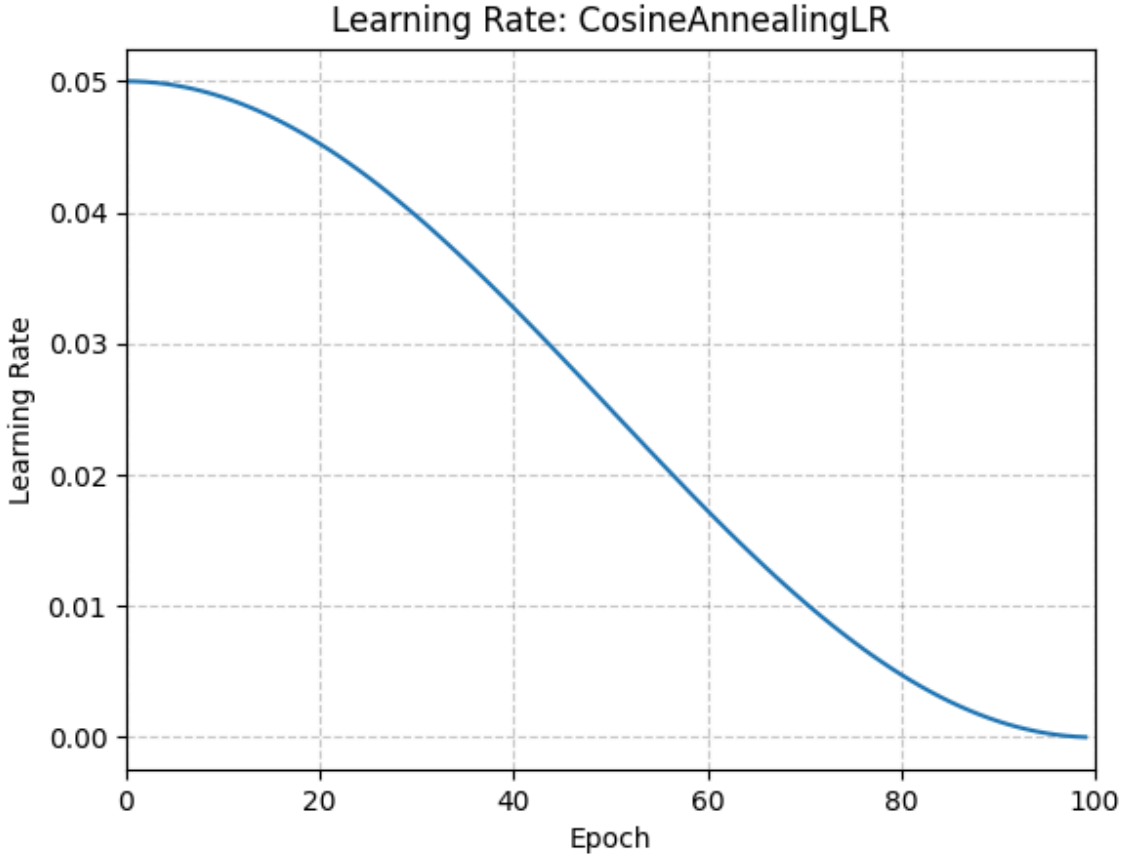
Hyperparameter	Value
β_1	0.9

β_2	0.999
ϵ	10^{-8}

Table 6.2. Fixed hyperparameters of the AdamW optimizer.

6.3.3 Learning Rate Scheduling

To improve convergence, a learning rate scheduling strategy is utilized. Two methods were considered initially, exponential and cosine schedulers, but early experiments showed the clear superiority of the cosine schedule, which was used throughout the entire experimentation process. The allowed range for the minimum value of the learning rate, η_{min} , which is a hyperparameter, is set to $[10^{-7}, 10^{-5}]$.



6.4 Experimental Design

6.4.1 Experiment 1: Best Hyperparameter Configuration

To find the best set of hyperparameters, 100 trials were run on the 10-second window in order to provide the model with as much information as possible. Each trial was given 500

epochs, with 150 epochs of no improvement set as the early-stopping threshold. The following configuration performed best and is therefore used in the remaining experiments:

Parameter	Value
Learning Rate	0.001368
Weight Decay	0.005068
Batch Size	128
η_{min}	0.0000086534

Table 6.3. Best hyperparameter configuration found by random search.

6.4.2 Experiment 2: Dependence on Window Length

The model is trained using different P-wave observation windows:

$$W \in \{3, 5, 8, 10\} \text{ seconds}$$

The main objective is to determine how the length of the early waveform window affects the performance of magnitude estimation. It is presumed that longer windows provide more information but may reduce the warning time. All results are evaluated on a held-out test set.

Window Length	Bias	MAE	Spearman r	R ²
3 s	−0.040	0.3699	0.7755	0.6483
5 s	−0.071	0.3453	0.8028	0.6851
8 s	−0.027	0.3217	0.8290	0.7225
10 s	−0.032	0.3152	0.8392	0.7478

Table 6.4. Performance of the proposed model as a function of the P-wave observation window length, evaluated on the test set. Best values for each metric are shown in bold.

Increasing the window length leads to improvement, though the difference is not large. This is expected because longer windows contain more information about the rupture. However, the improvement comes at the cost of increased warning latency. Thus, a trade-off exists between prediction accuracy and response time, which is critical for any EEW system.

6.5 Discussion

Several observations can be drawn from the results above.

First, the marginal gain in MAE when increasing the window from 5 to 10 seconds is approximately 0.03 magnitude units, while the gain from 3 to 5 seconds is roughly 0.025. This indicates a regime of diminishing returns: most of the magnitude-relevant information accessible to a single-station model is already present in the first few seconds of the P-wave. This implies that operationally useful estimates can be obtained from very short windows, at the cost of only a modest accuracy penalty.

Second, the coefficient of determination plateaus at $R^2 \approx 0.75$ for the longest window, meaning that approximately 25% of the variance in magnitude remains unexplained by the single-station model.

Third, the bias is small in absolute value (between -0.07 and -0.03). The tendency to slightly underestimate is likely caused by training on data with low-to-moderate magnitudes events prevalence. This pushes the need for either oversampling of high-magnitude events or synthetic data generation.

Fourth, Spearman's rank correlation reaches 0.84 for the 10-second window. Because this metric is sensitive to the ordering of predictions rather than their absolute values, it indicates that the model preserves the relative ranking of events well, even when individual point predictions are imperfect. For an alerting system that operates on a threshold rather than on point estimates, an argument for ranking quality importance being higher than absolute MAE can be made.

6.6 Visualization of Results

Several approaches are used to interpret the performance of the model. The predicted-versus-true scatter plot in Figure 6.1 provides an intuitive view of prediction on the test set. The corresponding residual histogram reveals that the model, on average, neither significantly overestimates nor underestimates the magnitude, with a mean residual of -0.032 .

Both graphs are extracted on the test set of 10s waveforms, using the best configuration as discussed before. The left depicts predicted magnitude plotted against ground-truth magnitudes, the perfect prediction reference is represented by a red dashed line. On the right, histogram of residuals (predicted - ground truth) is shown, with a mean of -0.032 .

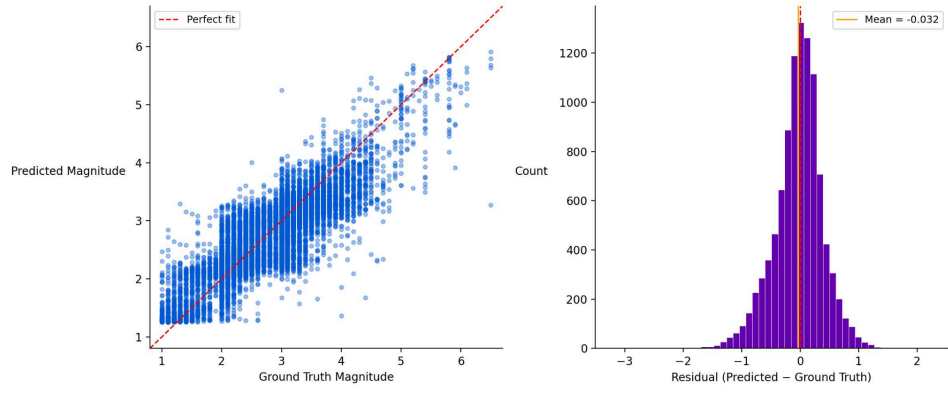


Figure 6.1

Figure 6.2 shows the training and validation loss alongside the MAE tracked over the course of training. Both training and validation curves decrease smoothly and stabilize at similar levels, indicating that the model converges without significant overfitting on the dense layers.

The graphs below depict the progression of both training and validation phases' loss and primary metric values. Visual similarity between training and validation phases indicates the stable training with the absence of overfitting.

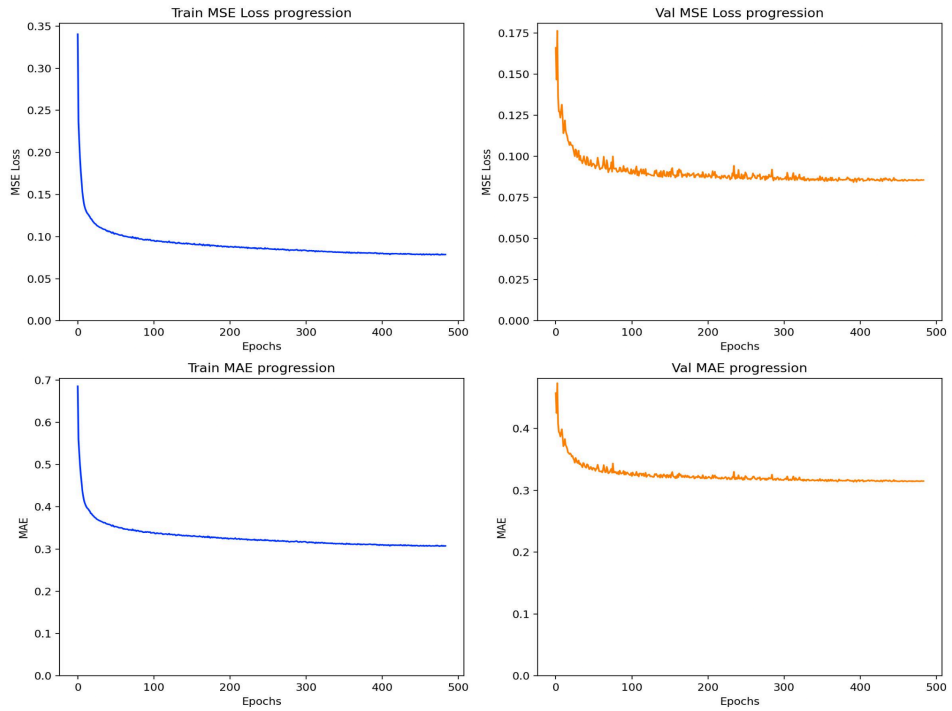


Figure 6.2.

6.7 Summary

This chapter described the experimental setting used to evaluate the proposed CNN-LSTM-MLP model for earthquake magnitude estimation, presented quantitative results

across multiple observation windows, and discussed the implications of these results for earthquake early warning systems.

CHAPTER 7. LIMITATIONS

7.1 Overview

This chapter discusses the limitations of the proposed approach from several perspectives, including dataset properties, methodological choices, and deployment considerations.

7.2 Data Imbalance

The most prominent limitation is the naturally inherited imbalance of the data. As discussed previously, the dataset is not rich in high-magnitude examples, and the model has therefore seen only a small fraction of large-magnitude events. While, in our case, we did not observe a significant divergence on the test set, the discrepancy between predictions and true magnitude values may diverge substantially for very large events ($M > 7.0$). Although such events are exceedingly rare, they are the ones that cause the most destruction. To address this problem, one can make use of carefully composed synthetic data to imitate large-magnitude events, however, such a technique requires deep expertise in both seismology and regional characteristics in order to minimize the risk of data corruption.

7.3 Single-Station Analysis

The proposed model operates on observations from a single station. Modern operational EEW systems typically aggregate observations from multiple stations to improve both accuracy and robustness. A single-station model is more vulnerable to local site effects and isolated noise events. Extending the proposed architecture to a multi-station setting is another direction for future work.

7.4 Dependence on P-wave Picking

The model relies on precise P-wave detection to extract the input window. While modern systems detect P-waves quite efficiently, small picking errors are still inevitable. In a real-world deployment, even small offsets in the assumed P-wave arrival can shift the input window and introduce additional noise to the input, which consequently increases uncertainty in the magnitude estimate. A robust deployment would either pair the model with a high-quality phase picker (such as PhaseNet) or expose the model to the data with imperfectly picked P-wave arrival time, this way the approach could potentially become more resilient to small timing errors.

7.5 Regional Variability

All training data come from the INSTANCE dataset, which is geographically concentrated in Italy. Seismic waveforms depend on local geological conditions. Consequently, a model trained on Italian seismicity is unlikely to perform well in a different region (for example, Kyrgyzstan) without fine-tuning or full retraining on local data.

7.6 Summary

This chapter discussed the main limitations of the proposed approach, including dataset imbalance, single-station design, dependence on P-wave picking, and regional variability of seismic signals. The results indicate that early seismic signals contain enough information to make an approximate estimate of the event magnitude, but several practical concerns must be addressed before such a system can be considered operationally mature.

CHAPTER 8. CONCLUSION

8.1 Summary of the Work

This thesis addressed the problem of earthquake magnitude estimation in the context of earthquake early warning systems. The main objective was to develop a deep learning model capable of predicting earthquake magnitude using only the first few seconds of seismic waveform data after P-wave arrival. To achieve this objective, the following steps were performed:

1. A review of existing machine learning approaches for earthquake early warning was conducted.
2. Mathematical foundations of signal processing and deep learning were established.
3. A preprocessing pipeline was developed to convert seismic waveforms into spectrogram representations.
4. A CNN–LSTM–MLP architecture was proposed for magnitude estimation.
5. An experimental framework was designed and used to evaluate model performance across multiple observation windows.

8.2 Key Findings

Based on the experimental results, the following conclusions can be drawn.

1. Early seismic signals contain meaningful information about earthquake magnitude. Using only the first 3 seconds of P-wave data, the proposed model achieves $MAE = 0.3699$ and $R^2 = 0.6483$, with 10 seconds of data, performance improves to $MAE = 0.3152$ and $R^2 = 0.7478$.
2. The combination of convolutional and recurrent neural networks is capable of modeling complex spatio-temporal data such as earthquake waveform spectrograms. Spearman's rank correlation reaches 0.84 for the 10-second window, indicating that the model preserves the ordering of magnitudes well across the test set.
3. Increasing the observation window improves accuracy but exhibits diminishing returns. Most of the magnitude-relevant information accessible to a single-station model is present in the first few seconds, suggesting a favorable trade-off between accuracy and warning latency.
4. PCA projection of the learned embeddings reveals a clear magnitude gradient along the first principal component, indicating that the CNN–LSTM backbone learns representations in which earthquake magnitude is one of the dominant axes of variation.

8.3 Contributions

The main contributions of this work are:

- the development of a spectrogram-based preprocessing pipeline for seismic data.
- the design of a hybrid CNN–LSTM–MLP architecture for early magnitude estimation.
- the integration of the INSTANCE seismic dataset into a unified training framework.
- the formulation of a structured experimental methodology for evaluating EEW models across multiple observation windows.
- an empirical demonstration that meaningful magnitude estimates can be obtained from as little as 3 seconds of single-station P-wave data.

8.4 Future Work

Several directions for future research can be identified.

In terms of architecture, potential improvements include the exploration of transformer-based models, more sophisticated attention mechanisms for temporal modeling, and lightweight architectures for faster inference.

In terms of data, future work may incorporate additional regional datasets and explore data enrichment via synthetic data generation techniques for larger earthquakes, with appropriate modeling to avoid data contamination.

At the software level, future development could involve model quantization to improve inference speed on micro-controllers that lack a dedicated graphics processing unit and can perform only integer-based arithmetic.

Methodologically, comparing the proposed model against a multi-station baseline and against established prior approaches (such as MagNet and TEAM) on a common evaluation protocol would strengthen the empirical case for spectrogram-based architectures in EEW.

8.5 Final Remarks

Earthquake early warning remains a challenging problem due to the inherent uncertainty and variability of seismic processes. While deep learning approaches cannot solve this problem outright, there is significant unexplored territory at the intersection of deep learning and seismology. The results presented in this thesis suggest that spectrogram-based hybrid architectures represent a productive line of investigation for this field.

REFERENCES

- [1] Bommer J. J. Earthquake hazard and risk analysis for natural and induced seismicity: towards objective assessments in the face of uncertainty // *Bulletin of Earthquake Engineering*. – 2022. – V. 20, N. 6. – P. 2825–3069.
- [2] Shearer P. M. *Introduction to Seismology*. – 2nd ed. – Cambridge: Cambridge University Press, 2009. – 412 p.
- [3] Allen R. M., Melgar D. Earthquake Early Warning: Advances, Scientific Challenges, and Societal Needs // *Annual Review of Earth and Planetary Sciences*. – 2019. – V. 47. – P. 361–388.
- [4] Minson S. E., Meier M.-A., Baltay A. S., Hanks T. C., Cochran E. S. The limits of earthquake early warning: Timeliness of ground motion estimates // *Science Advances*. – 2018. – V. 4, N. 3. – eaaq0504.
- [5] Mousavi S. M., Beroza G. C. Deep-learning seismology // *Science*. – 2022. – V. 377, N. 6607. – eabm4470.
- [6] Aki K., Richards P. G. *Quantitative Seismology*. – 2nd ed. – Sausalito, CA: University Science Books, 2002. – 700 p.
- [7] Meier M.-A., Heaton T., Clinton J. Evidence for universal earthquake rupture initiation behavior // *Geophysical Research Letters*. – 2016. – V. 43, N. 15. – P. 7991–7996.
- [8] Allen R. V. Automatic earthquake recognition and timing from single traces // *Bulletin of the Seismological Society of America*. – 1978. – V. 68, N. 5. – P. 1521–1532.
- [9] Zhu W., Beroza G. C. PhaseNet: a deep-neural-network-based seismic arrival-time picking method // *Geophysical Journal International*. – 2019. – V. 216, N. 1. – P. 261–273.
- [10] Mousavi S. M., Ellsworth W. L., Zhu W., Chuang L. Y., Beroza G. C. Earthquake transformer — an attentive deep-learning model for simultaneous earthquake detection and phase picking // *Nature Communications*. – 2020. – V. 11. – 3952.
- [11] Michelini A., Cianetti S., Gaviano S., Giunchi C., Jozinović D., Lauciani V. INSTANCE – the Italian seismic dataset for machine learning // *Earth System Science Data*. – 2021. – V. 13. – P. 5509–5544.
- [12] Gutenberg B., Richter C. F. Frequency of earthquakes in California // *Bulletin of the Seismological Society of America*. – 1944. – V. 34, N. 4. – P. 185–188.

- [13] Scholz C. H. The Mechanics of Earthquakes and Faulting. – 3rd ed. – Cambridge: Cambridge University Press, 2019. – 519 p.
- [14] Hanks T. C., Kanamori H. A moment magnitude scale // Journal of Geophysical Research: Solid Earth. – 1979. – V. 84, N. B5. – P. 2348–2350.
- [15] Ruhl C. J., Melgar D., Grapenthin R., Allen R. M. The value of real-time GNSS to earthquake early warning // Geophysical Research Letters. – 2017. – V. 44, N. 16. – P. 8311–8319.
- [16] Mousavi S. M., Zhu W., Sheng Y., Beroza G. C. CRED: A deep residual network of convolutional and recurrent units for earthquake signal detection // Scientific Reports. – 2019. – V. 9. – 10267.
- [17] Lin J.-T., Melgar D., Thomas A. M., Searcy J. Early warning for great earthquakes from characterization of crustal deformation patterns with deep learning // Journal of Geophysical Research: Solid Earth. – 2021. – V. 126, N. 10. – e2021JB022703.
- [18] Münchmeyer J., Bindi D., Leser U., Tilmann F. The transformer earthquake alerting model: a new versatile approach to earthquake early warning // Geophysical Journal International. – 2021. – V. 225, N. 1. – P. 646–656.
- [19] Mousavi S. M., Beroza G. C. A machine-learning approach for earthquake magnitude estimation // Geophysical Research Letters. – 2020. – V. 47, N. 1. – e2019GL085976.
- [20] Zhu W., Mousavi S. M., Beroza G. C. Seismic signal denoising and decomposition using deep neural networks // IEEE Transactions on Geoscience and Remote Sensing. – 2019. – V. 57, N. 11. – P. 9476–9488.
- [21] LeCun Y., Bengio Y., Hinton G. Deep learning // Nature. – 2015. – V. 521, N. 7553. – P. 436–444.
- [22] Ioffe S., Szegedy C. Batch normalization: accelerating deep network training by reducing internal covariate shift // Proceedings of the 32nd International Conference on Machine Learning (ICML). – Lille, France. – 2015. – P. 448–456.
- [23] Srivastava N., Hinton G., Krizhevsky A., Sutskever I., Salakhutdinov R. Dropout: a simple way to prevent neural networks from overfitting // Journal of Machine Learning Research. – 2014. – V. 15. – P. 1929–1958.
- [24] Hochreiter S., Schmidhuber J. Long short-term memory // Neural Computation. – 1997. – V. 9, N. 8. – P. 1735–1780.

- [25] Loshchilov I., Hutter F. Decoupled weight decay regularization // Proceedings of the 7th International Conference on Learning Representations (ICLR). – New Orleans, LA, USA. – 2019.
- [26] Bergstra J., Bengio Y. Random search for hyper-parameter optimization // Journal of Machine Learning Research. – 2012. – V. 13. – P. 281–305.